

LE-OFTs are also expected to become fully compatible with well-established electronic technologies. This may further allow the development of optical communication systems and opto-electronic systems.

LE-OFTs are being used to develop flexible, transparent computer screens, which are used to provide faster response times, better efficiency, and these do not need backlighting. These also have very low energy needs.

Light-receiving OFETs, on the other hand, are much less developed than their light-emitting counterparts.

Internet of Things sensors can run at lower power

As more sensors get added to the Internet of Things (IoT), power consumption may pose a problem. However, researchers at Massachusetts Institute of Technology, the USA, have come up with a solution to reduce the energy that such components require.

They have created a new circuit design that could lead to more power-efficient sensors, which are driving growth in the IoT segment. About 1.2 billion IoT-connected devices receive or transmit data wirelessly for alerts or analysis, and this number could grow to 5.4 billion by 2020, as per a study by Verizon.

Many sensors remain idle most of the time and become active when these send or receive data. Researchers have designed a circuit for a transmitter that could reduce energy leakage by up to 100 times when a sensor is in idle state. This could extend the battery life of sensors by months.

Wireless technologies are becoming more power-efficient but can still be one of the biggest electricity hogs in small devices. The new design does not compromise on wireless connectivity; it can generate enough power to transmit data using Bluetooth or 802.15.4 specification, which is the basis for Zigbee protocol.

The new circuit reduces transistor leakage by passing a negative charge through a set of capacitors and switches during idle time. The negative charge effectively banishes electrons from electrical leads, thereby reducing the conductivity of the silicon and also the opportunity for leakage.

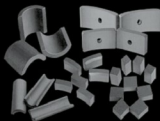
The team at Massachusetts Institute of Technology tested a chip based on the new design and found that it used only 20 picowatts of power to save 10,000 picowatts that would otherwise have been lost to leakage.

The research was led by Anantha Chandrakasan, professor of computer science at the institute. He has also researched technology that could self-power IoT devices by drawing energy from body heat, solar energy or ambient surroundings. The new transmitter design could complement the energy-harvesting technologies to make IoT devices more power-efficient.

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